ACTIVITY 2

Find out major organs that are part of the circulatory system of a human body and listout their functions

4.2 Plant and Animal cell comparison

Why do plant cells differ from animal cells? They differ from each other because they have to perform different functions.

Now you know that there are many main similarities between plant and animal cells. Let us see how they differ from one another as given in the picture (Activity 3).

4.3 Human cells related to functions

Different types of cells

Our body is made up of many different kinds of cells. Each type of cell is specialized to perform a specific function. Depending on the function, cell has specific shape, size and may have some components which other type of cells do not have. Have a look at the differences between nerve cells and red blood cells in the images. Even though there are many different types of cells, there are some components common to all type of cells. Let us take a look at this in the next section.

What's inside a cell?

Inside a cell, there are many tiny structures called cell organelles. These organelles are responsible for providing needs of the cell. They work to bring in food supplies, get rid of waste, protection and repair of the cell, and help it to grow and reproduce. Each one has a specific function to do for the cell. And, if any one organelle stops its function, then the cell is programmed to die.

Cell Structure

As we have mentioned before, all cells have some common structure.

These are

1. Cell membrane
2. Cytoplasm, and
3. Nucleus (In most eukaryotic cells).

The structure of a typical plant and animal cell shows following peculiarities:

Cell membrane

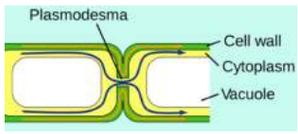
The boundary of an animal cell is the plasma membrane, which is also called as cell membrane

Cell wall - "Supporter and Protector"

All animal and plant cells are enclosed or surrounded by a cell membrane as you learned before. However, as you might have noticed previously that, animal cells often have an irregular shape, whereas plant cells have a much more regular and rigid shape.

Plant cells have an additional layer on the outer side of the cell membrane. This is called as the cell wall, that provides a frame work for support and stability.

The cell wall is formed from various compounds, the main one being cellulose. Cellulose helps to maintain the shape of the plant cell. This allows the plant to remain rigid and upright even if it grows to great heights. Each cell is interconnected with its neighboring cells through openings called Plasmodesmata.



Cell wall

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